



**STEEL AUTHORITY OF INDIA LIMITED
BHILAI STEEL PLANT
RESEARCH & CONTROL LAB (MECHANICAL TESTING LAB)**

Page No :B-1 of 1

IS : 2062



CM/L-0057534
MTL-II/QR-2

TEST CERTIFICATE FOR HOT ROLLED MEDIUM AND HIGH TENSILE STRUCTURAL STEEL

Sold-to-Party:

BM BSO JAIPUR

S-16A LANDMARK 4TH FLOOR OPP:GUJARATI SAMAJ MAHAVIR

JAIPUR 302001

Ship-to-Party:

WHM JAIPUR,Jaipur.

T.C.No

: RCL/MTL/PLM/80927599

T.C. Date

: 04.03.2026

SALES ORDER NO

: 1100584228

DA No

: **80927599**

Wagon No / Trailer No

: ECOR94121336513

Process of Manufacture of Steel : Basic Oxygen Converter> Continuous Cast> Fully Killed Steel

Specification : IS 2062 E 350 BR : 2011 RA 2021

We certify that the material described below fully conforms to IS 2062 : 2011 RA 2021. Chemical composition and Mechanical properties of the product, as tested in accordance with the scheme of testing and inspection contained in the BIS certification mark Number CM/L 0057534 are as indicated against each Order No.

(Please refer to IS 2062 : 2011 RA 2021 for details of specification requirements)

TEST RESULTS

SECTION (NOMINAL SIZE)			Plate No / Pc. No.	Heat No	MECHANICAL PROPERTIES										Bend Test	M Dia
THICK	WIDTH	LENGTH			YS	UTS	%E	%E2	CHARPY V NOTCH (L) Impact Values (J)							
mm	mm	mm			330 MPa Min	490 MPa	Min22(5.6 5SqrtS0)		I1	I2	I3	Iavg	°C			
40	2000	12000	3493928/1	411499	423	555	24	--	--	--	--	RT	--	--		
				411499	414	529	25	--	110	114	122	115	RT	--	--	
40	2000	12000	3493929/1	411499	--	--	--	--	--	--	--	--	--	--	--	
40	2000	12000	3493930/1	411499	--	--	--	--	--	--	--	--	--	--	--	
40	2000	12000	3494904/1	411501	439	564	23	--	--	--	--	--	RT	--	--	
				411501	407	548	24	--	112	106	108	109	RT	--	--	
40	2000	12000	3494908/1	411501	--	--	--	--	--	--	--	--	--	--	--	
40	2000	12000	3494909/1	411501	--	--	--	--	--	--	--	--	--	--	--	
Heat No	CHEMICAL COMPOSITION (LADLE ANALYSIS)															
	C	S	P	Mn	Si	Al	Cu	Cr	Ni	Mo	Nb	V	Ti	N ₂	B	CE
	%	%	%	%	%	%	%	%	%	%	%	%	%	ppm	%	%
411499	0.16	0.013	0.021	1.44	0.32	0.024	< 0.02	< 0.015	< 0.015	< 0.015	0.033	< 0.010	< 0.005	59	< 0.0005	0.40
411501	0.17	0.013	0.020	1.42	0.30	0.023	< 0.02	< 0.015	< 0.015	< 0.015	0.031	< 0.010	< 0.005	61	< 0.0005	0.41

Total Pieces Page / Sales Order : 6 / 6

Total Pieces Page / Sales Order : 6 / 6

Total Pieces Page / Sales Order : 6 / 6

Total Pieces / TC

: 8

The Material supplied conforms to the standard rolling and weight tolerances.,

CE calculated by formula: $[C + Mn/6 + (Cr + Mo + V)/5 + (Ni + Cu)/15]$,

RT- Room temperature as defined in the specification,

Size of impact test specimen: 10x10x55 mm(above 10mm thickness) ; 7.5x10x55 mm(upto 10mm thickness),

Note:** M = Mandrel Dia, T = Plate Thickness, Length is the rolling direction of the plate,

TC as per BSEN 10204 Type 3.1,

Signature valid

Digitally signed by S. BAIJU
Date: 2026.03.04 19:23:44 IST
Reason: Authorized Signatory RCL,
BSP

Authorised Signatory
SAIL/ CMO/ BSO

Authorized Signatory
Research & Control Laboratory, Bhilai Steel Plant



**STEEL AUTHORITY OF INDIA LIMITED
BHILAI STEEL PLANT
RESEARCH & CONTROL LAB (MECHANICAL TESTING LAB)**

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IS : 2062



CM/L-0057534
MTL-II/QR-2

**TEST CERTIFICATE FOR HOT ROLLED MEDIUM AND HIGH TENSILE STRUCTURAL
STEEL**

Sold-to-Party:

BM BSO JAIPUR

S-16A LANDMARK 4TH FLOOR OPP:GUJARATI SAMAJ MAHAVIR

JAIPUR 302001

Ship-to-Party:

WHM JAIPUR,Jaipur.

T.C.No : RCL/MTL/PLM/80927599

T.C. Date : 04.03.2026

SALES ORDER NO : 1100584205

DA No : **80927599**

Wagon No / Trailer No : ECR94121336513

Process of Manufacture of Steel : Basic Oxygen Converter> Continuous Cast> Fully Killed Steel

Specification : IS 2062 E 250 BR : 2011 RA 2021

We certify that the material described below fully conforms to IS 2062 : 2011 RA 2021. Chemical composition and Mechanical properties of the product, as tested in accordance with the scheme of testing and inspection contained in the BIS certification mark Number CM/L 0057534 are as indicated against each Order No.

(Please refer to IS 2062 : 2011 RA 2021 for details of specification requirements)

TEST RESULTS

SECTION (NOMINAL SIZE)			Plate No / Pc. No.	Heat No	MECHANICAL PROPERTIES										Bend Test	M Dia
THICK	WIDTH	LENGTH			YS	UTS	%E	%E2	CHARPY V NOTCH (L) Impact Values (J)							
mm	mm	mm			240 MPa Min	410 MPa Min	Min23(5.6 5SqrtS0)		I1	I2	I3	Iavg	°C			
40	2000	12000	3502109/1	412240	327	467	26	--	--	--	--	--	RT	OK	3.0 T	
				412240	334	474	25	--	66	64	68	66	RT	OK	3.0 T	
40	2000	12000	3504143/1	412407	331	478	25	--	--	--	--	--	RT	OK	3.0 T	
				412407	328	475	26	--	90	86	84	87	RT	OK	3.0 T	
Heat No	CHEMICAL COMPOSITION (LADLE ANALYSIS)															
	C	S	P	Mn	Si	Al	Cu	Cr	Ni	Mo	Nb	V	Ti	N ₂	B	CE
	%	%	%	%	%	%	%	%	%	%	%	%	%	ppm	%	%
412240	0.19	0.019	0.035	0.95	0.28	0.015	< 0.02	< 0.015	< 0.015	< 0.015	< 0.010	< 0.010	< 0.005	--	< 0.0005	0.35
412407	0.18	0.018	0.025	0.95	0.30	0.030	< 0.02	< 0.015	0.020	< 0.015	< 0.010	< 0.010	< 0.005	--	< 0.0005	0.34

Total Pieces Page / Sales Order : 2 / 2

Total Pieces / TC

: 8

The Material supplied conforms to the standard rolling and weight tolerances.,

CE calculated by formula: $[C + Mn/6 + (Cr + Mo + V)/5 + (Ni + Cu)/15]$,

RT- Room temperature as defined in the specification,

Plate sides are in UNTRIMMED condition.,

Size of impact test specimen:10x10x55 mm(above 10mm thickness); 7.5x10x55 mm(upto 10mm thickness),

Note:** M = Mandrel Dia, T = Plate Thickness, Length is the rolling direction of the plate,

TC as per BSEN 10204 Type 3.1,

Signature valid

Digitally signed by S. S. JAIPUR
Date: 2026.03.04 17:23:44 IST
Reason: Authorized Signatory RCL, BSP

Authorised Signatory
SAIL/ CMO/ BSO

Authorized Signatory
Research & Control Laboratory, Bhilai Steel Plant